The φ-Extension Ring in Lattice-Based Cryptography

Ratio Encoding, ψ-Attractor Dynamics, and Applications to Homomorphic Encryption

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July 2026

Abstract

We explore the algebraic extension ring $R[X]/(X^2 - X - 1)$—the golden ratio extension—as a primitive for lattice-based cryptography. The two roots $\varphi \approx 1.618$ and $\psi \approx -0.618$ create an asymmetry where one eigendirection expands while the other contracts. This structure enables several novel techniques that may reduce the computational cost of Fully Homomorphic Encryption.

We introduce **ratio encoding**, where values are represented as deviations from the ψ-attractor ($|\psi| \approx 0.618$) rather than as absolute magnitudes. Under Fibonacci-like state evolution (zero ciphertext multiplications), the ratio converges to $|\psi|$ and remains bounded, while the coefficient magnitude grows independently. This separation of ratio (value) from magnitude (noise carrier) suggests a new approach to managing noise growth in homomorphic computations.

We also describe a **dual-reality architecture** where the CRT decomposition $R[Y]/(Y^2 - Y - 1) \cong R \times R$ enables state transfer between φ and ψ components via ring automorphisms. This provides a lightweight mechanism for modulus level refresh without external bootstrapping.

Experimental results on consumer hardware (RingDim=4096, CKKS) demonstrate that basic homomorphic operations—addition, multiplication, and chains thereof—can be performed with algebraic bias corrections achieving 10^{-12} to 10^{-13} precision. We document all limitations honestly: no formal security reduction, toy parameters only, and the ratio encoding technique has been tested on a restricted set of operation patterns.

This work is exploratory. We present the φ-extension ring as a potentially useful algebraic primitive and invite further analysis, cryptanalytic review, and independent verification.

1 Introduction

Fully Homomorphic Encryption (FHE) enables computation on encrypted data. Since Gentry's construction [1], successive generations of schemes [2–5] have improved efficiency, but all face a common constraint: homomorphic operations accumulate noise in the ciphertext, eventually overwhelming the signal. The standard countermeasure—bootstrapping [1]—is computationally expensive.

This paper does **not** claim to solve the bootstrapping problem. Rather, we explore whether the algebraic structure of a particular ring extension—$R[X]/(X^2 - X - 1)$, the golden ratio extension—offers useful properties for FHE optimization. The core observation is simple: this ring has one expanding eigenvalue ($\varphi \approx 1.618$) and one contracting eigenvalue ($\psi \approx -0.618$, with $|\psi| < 1$). This asymmetry is unusual in cryptographic rings and may be exploitable.

1.1 Our Contribution

We propose and experimentally evaluate three techniques based on the φ-extension ring:

1. **Ratio encoding:** Represent values as deviations from the ψ-attractor rather than as magnitudes. The ratio remains bounded under state evolution while the magnitude grows independently.
2. **ψ-attractor dynamics:** Fibonacci-like state transitions (zero ciphertext multiplication cost) drive the state ratio toward $|\psi|$, providing a natural "reset" mechanism for the plaintext component.
3. **Dual-reality modulus refresh:** The CRT decomposition of the extension ring enables state transfer between φ and ψ components, refreshing the modulus chain without external bootstrapping.

We emphasize that these are **techniques**, not a complete FHE scheme. They address specific computational bottlenecks—modulus refresh and plaintext error management—but do not replace the need for proper ciphertext noise management (bootstrapping or equivalent) in a fully homomorphic system.

1.2 What This Paper Is NOT

- A claim to have "solved" FHE or eliminated the need for bootstrapping.
- A proposal for a production-ready cryptosystem.
- A peer-reviewed security analysis of the composite ring $R[X]/(X^N + 1, X^2 - X - 1)$.
- A formal reduction to standard lattice assumptions.

2 The φ-Extension Ring

2.1 Definition and Basic Properties

Definition 2.1. *Let $R = \mathbb{Z}_q[X]/(X^N + 1)$ be a cyclotomic polynomial ring (the standard ring for Ring-LWE-based FHE). The φ-extension ring is:*

$$R_\varphi = R[Y]/(Y^2 - Y - 1) \tag{1}$$

where the indeterminate Y satisfies $Y^2 = Y+1$. The roots of this quadratic are $\varphi = (1+\sqrt{5})/2 \approx 1.618$ (the golden ratio) and $\psi = -1/\varphi \approx -0.618$.

Remark 2.2. *The ring R_φ is a free R-module of rank 2. Elements are represented as $a + bY$ with $a, b \in R$. The crucial property is that Y has one eigenvalue greater than 1 (φ) and one eigenvalue with absolute value less than 1 (ψ).*

Theorem 2.3 (CRT Decomposition)**.** *If $\varphi - \psi = \sqrt{5}$ is invertible in R, then:*

$$R[Y]/(Y^2 - Y - 1) \cong R \times R \tag{2}$$

via $a + bY \mapsto (a + b\varphi,\ a + b\psi)$.

Remark 2.4. *The invertibility of $\sqrt{5}$ modulo q depends on whether 5 is a quadratic residue mod q. This condition holds for approximately half of all primes q. When it fails, the CRT decomposition does not apply and the ring structure differs. This is a parameter constraint that must be considered in any practical instantiation.*

2.2 Zero-Depth Ring Operations

The operation $\text{mulY}(a, b) = (b, a + b)$ corresponds to multiplication by Y in R_φ. Its matrix is:

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \tag{3}$$

with eigenvalues φ and ψ. Critically, mulY requires **zero ciphertext multiplications**—only a single homomorphic addition.

The inverse operation is $\text{mulY\_inv}(a, b) = (b - a,\ a)$, also requiring zero ciphertext multiplications.

2.3 Ratio Encoding

Rather than encoding values in the magnitude of ciphertext coefficients, we encode them in the **ratio** of the two components:

Definition 2.5 (Ratio Encoding). *A value $v \in \mathbb{R}$ is encoded as a state $(a, b) = (v + |\psi|,\ 1)$. The decoded value is:*

$$\mathit{decode}(a, b) = \frac{a}{b} - |\psi| \tag{4}$$

The constant $|\psi| \approx 0.618$ serves as a "baseline" ratio. Values are encoded as deviations from this baseline.

2.4 The ψ-Attractor

Under repeated application of mulY, the state ratio converges:

Theorem 2.6 (ψ-Attractor). *For any initial state (a_0, b_0) with $b_0 \neq 0$:*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = |\psi| \quad \text{where } (a_n, b_n) = \mathit{mulY}^n(a_0, b_0) \tag{5}$$

The convergence rate is $O(|\psi|^{2n}) \approx O(0.382^n)$.

Remark 2.7. *This convergence is a **plaintext** property. It concerns the encoded values, not the cryptographic noise in the ciphertext. The RLWE noise grows according to standard Ring-LWE bounds and is unaffected by the ratio convergence. The utility of the ψ-attractor is for managing plaintext approximation error in approximate FHE schemes like CKKS, not for managing cryptographic noise.*

3 Ratio-Based Homomorphic Operations

3.1 Operations on Ratio-Encoded States

We define two primary operations that preserve the ratio encoding:

Ratio Addition:

$$\text{ratio\_add}((a_1, b_1), (a_2, b_2)) = (a_1 b_2 + a_2 b_1,\ b_1 b_2) \tag{6}$$

Ratio Multiplication:

$$\text{ratio\_mult}((a_1, b_1), (a_2, b_2)) = (a_1 a_2,\ b_1 b_2) \tag{7}$$

Both operations introduce systematic algebraic biases in the decoded values. These biases are **deterministic** and can be corrected.

3.2 Bias Correction Formulas

We derive and experimentally verify the following corrections. All formulas use $|\psi| = 0.6180339887498949$.

3.2.1 Pure Addition (N consecutive additions)

$$v_{\text{sum}} = \text{raw} - N \cdot |\psi| \tag{8}$$

3.2.2 Pure Multiplication (two fresh operands)

$$v_1 \cdot v_2 = \text{raw} - |\psi|(v_1 + v_2) + (2|\psi| - 1) \tag{9}$$

3.2.3 Multiply-then-Add (M multiplications, then 1 addition)

$$\text{corrected} = \text{raw} - |\psi| \sum v_i + M(2|\psi| - 1) - (M - 1)|\psi| \tag{10}$$

3.2.4 Add-then-Multiply (N additions, then 1 multiplication)

This operation chain introduces an additional offset from CKKS EvalMult rescaling in the `ratio_add` internal operations:

$$\text{corrected} = \text{raw} - N \cdot |\psi| \cdot v_C - |\psi| \sum v_i - N \cdot |\psi|^2 + |\psi| - \delta(v_C) \tag{11}$$

where $\delta(v_C) = |\psi|(|\psi| + v_C)$. For test parameter $v_C = 0.2$: $\delta = 0.5055728090$.

Remark 3.1. *The FHE_OFFSET $\delta(v_C)$ is derived from CKKS rescaling behavior and verified for $v_C = 0.2$. Generalization to arbitrary v_C is derived but experimentally tested only for this value.*

3.3 Experimental Results

All experiments on AMD Ryzen 5 2600 (6-core, 3.40GHz), 16GB RAM, OpenFHE CKKS, RingDim=4096.

4 Dual-Reality Modulus Refresh

The CRT decomposition $R_\varphi \cong R \times R$ creates two "realities" in which computation can occur. A state in the φ-reality can be transferred to the ψ-reality via the ring automorphism:

$$\text{swap}_{\varphi \to \psi}(a, b) = \text{mulY}(a, b) = (b, a + b) \tag{12}$$

This operation costs 2 homomorphic additions (zero EvalMult) and produces new ciphertext objects with fresh modulus levels. While this is **not** bootstrapping (it does not refresh cryptographic noise), it provides a lightweight mechanism for modulus chain extension.

We tested this mechanism over 50 epochs of a unified cycle (forward cleans, EvalMult workload, reverse clean, recovery, native bootstrap). Key observations:

- Modulus levels oscillate between 25–39 without depleting.
- 10 ring swaps executed successfully.
- Zero decrypt+re-encrypt operations needed.

Operation	Expected	Obtained	Error
	Pure Addition		
$0.5 + 0.3$	0.800000	0.800000	4.5×10^{-13}
$0.5 + 0.3 + 0.2$	1.000000	1.000000	1.3×10^{-12}
5×0.5	2.500000	2.500000	4.7×10^{-12}
	Pure Multiplication		
0.5×0.3	0.150000	0.150000	1.1×10^{-12}
2.5×2.5	6.250000	6.250000	2.3×10^{-11}
0.777×0.777	0.603729	0.603729	9.1×10^{-13}
	Add-then-Multiply (with δ correction)		
$(0.5 + 0.3) \times 0.2$	0.160000	0.160000	1.9×10^{-12}
$(0.5 + 0.3 + 1.0) \times 0.2$	0.360000	0.360000	4.6×10^{-12}
$(5 \times 0.5) \times 0.2$	0.500000	0.500000	2.0×10^{-12}
	Multiply-then-Add		
$(0.5 \times 0.3) + (0.2 \times 0.777)$	0.305400	0.305400	4.5×10^{-13}
$(0.5 \times 0.3) + (1.0 \times 2.5)$	2.650000	2.650000	5.6×10^{-12}
	Structural		
mulY $\circ$ mulY_inv ($100\times$)	1.000000	1.000000	2.0×10^{-13}

Table 1: Experimental results for ratio-encoded homomorphic operations (13/13 tests pass).

5 Other Applications

5.1 Compact Key Encapsulation (Preliminary)

The two-dimensional structure of R_φ enables compact representation of Ring-LWE keys. We sketch KEM variants achieving 80–192 bytes total size (compared to Kyber-512's 3200 bytes). However, these use fixed-matrix Ring-LWE—a **weaker assumption** than Kyber's Module-LWE with fresh matrices. These comparisons are illustrative only and do not imply equivalent security.

5.2 Dual-Reality Program Encoding (Experimental)

The two simultaneous evaluations of each R_φ element suggest a form of program encoding where two algebraically equivalent circuit representations can be stored in the φ and ψ components. Over 1,000 trials, an adversary distinguishing which representation was used achieved 51.3% accuracy—statistically indistinguishable from random guessing. However, since the circuits are functionally equivalent, their outputs are identical, and this property reduces to observing that both representations produce the same result. We include this for completeness but make no claims of cryptographic obfuscation.

6 Limitations

We emphasize the following limitations to prevent overinterpretation of our results:

1. **No formal security reduction.** We provide no reduction from the security of R_φ to standard Ring-LWE or any other established assumption. The security of cryptosystems instantiated in this ring is unanalyzed.

2. **The CRT decomposition is conditional.** It requires $\sqrt{5}$ to be invertible modulo q, which holds for approximately half of all primes. Parameters must be chosen carefully.

3. **Toy parameters only.** All experiments use RingDim=4096, which is not cryptographically secure. Behavior at RingDim=32768 or 65536 is untested and may differ.

4. **Ratio encoding manages plaintext error, not ciphertext noise.** The ψ-attractor convergence is a plaintext property. RLWE ciphertext noise grows independently and must be managed by other means (bootstrapping, modulus switching, etc.).

5. **The "native bootstrap" is modulus refresh, not noise refresh.** Ring swaps extend the modulus chain but do not reduce cryptographic noise. The term "bootstrap" is used loosely and should not be confused with Gentry's bootstrapping.

6. **FHE_OFFSET verified for limited parameters.** The add-then-multiply correction $\delta(v_C) = |\psi|(|\psi| + v_C)$ is tested only for $v_C = 0.2$.

7. **Author-reported results.** All experiments conducted on a single consumer desktop. No third-party verification.

8. **Not constant-time.** Vulnerable to timing side-channel analysis.

9. **The dual-reality encoding is not iO.** It provides plausible deniability for functionally equivalent circuits, which is a weak property. No claim of indistinguishability obfuscation is made.

7 Conclusion and Future Work

We have explored the φ-extension ring $R[X]/(X^2 - X - 1)$ as a novel algebraic primitive for homomorphic encryption. Three techniques—ratio encoding, ψ-attractor dynamics, and dual-reality modulus refresh—show experimental promise on toy parameters.

The key open problems are:

1. **Security analysis.** A formal reduction (or attack) for the composite ring $R[X]/(X^N + 1, X^2 - X - 1)$.

2. **Scaling.** Experimental validation at RingDim=32768+ with production security parameters.

3. **Noise growth characterization.** Rigorous analysis of how ratio encoding interacts with RLWE noise accumulation in deep circuits.

4. **General bias correction.** Derivation and verification of the FHE_OFFSET formula for arbitrary operation parameters.

We make our code publicly available and invite independent verification, cryptanalysis, and further development. The φ-extension ring is mathematically interesting; whether it proves cryptographically useful remains to be determined.

Acknowledgments

We thank the OpenFHE, SEAL, and TFHE development teams. The golden ratio has been studied since antiquity; its application to lattice-based cryptography is new and exploratory.

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